

in 2007, SDPs are built on a need-to-know model with trust that is constantly monitored and adapted based on a range of criteria. They make application infrastructure invisible to the internet, reducing the attack surface for network based attacks (DDoS, ransomware, malware, server scanning, etc). SDP guarantees that all endpoints attempting to access enterprise infrastructure are authenticated and approved.

This perimeter takes a different approach compared to traditional network based security. Instead of focusing on securing the network, it focuses on securing the user, the application, and the connectivity in between. Four core principles differentiate SDP technologies.

1. **Trust is never implicit:** SDPs only grant application access to users who are authenticated and specifically authorised to use an app. Furthermore, authorised users are only granted access to the application, not the network.
2. **No inbound connections:** By responding with outbound-only connections, SDPs keep the network and application infrastructure invisible to the internet and therefore impossible to attack.
3. **Application segmentation, not network segmentation:** SDP provides native application segmentation that can control access on a one-to-one basis, resulting in far more granular segmentation, which is much easier for IT teams to manage.

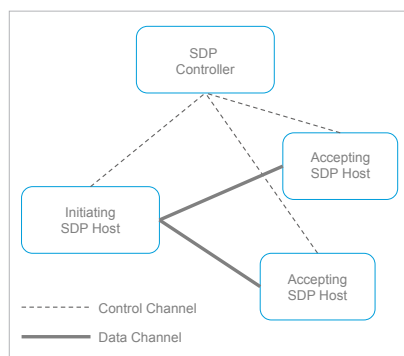


Figure 1: Software defined perimeter modules

4. **Secure internet leverage:** SDP is focused on securing user-to-application connections over the internet rather than securing users' access to the network.

Figure 1 shows SDP components and interactions. SDP architectures are made up of two main components — SDP controllers and SDP hosts. Control and data channels are separated. An SDP controller determines which SDP hosts can communicate with each other. An SDP host can be either initiating or accepting. An initiating SDP host communicates with an SDP controller to determine which hosts they can connect to. An accepting SDP host only accepts allowed communications and connections from an SDP controller.

Zero trust and SDP are based on the same philosophy. Rooted in the principle of 'never trust, always verify', they are designed to protect distributed digital environments by leveraging segmentation, preventing lateral movement, providing a Layer 7 threat prevention, and simplifying granular user-access control.

An SDP not only reduces the risk to endpoints that connect from anywhere, but also—with the help of an SDP controller—distributes network resources more evenly. Because these resources are defined on an individual basis, access control is centralised as well as simplified, ensuring secure access throughout the entire organisation. Thus, SDP

addresses enterprise issues by enabling remote access use cases such as finding VPN alternatives, securing multi-cloud access, and reducing third-party risks.

SDP or zero trust network architecture denies access to resources unless the user or machine has been given explicit permission. There are no implicit trust relationships. Moreover, those access rights are continually evaluated and approved (or declined) in real-time for each identity, every time access is requested.

This 'never trust, always verify' validation policy is the primary difference between zero trust and legacy network security models. It works through a variety of methods, including user authentication, authorisation, and inspection, and is based on criteria, such as a user's identity, location, operating system and firmware version, and endpoint hardware type. The zero trust approach provides granular, least-privilege access to limit lateral movement.

Blockchain defined perimeter

SDP and zero trust platforms provide comprehensive secure access to applications and environments, independent of user, device, and location. SDP, stitched together with blockchain technology, is able to deliver fine-grained micro-segmented network access to business applications, irrespective of whether they are hosted on-premises or across one or multiple clouds.

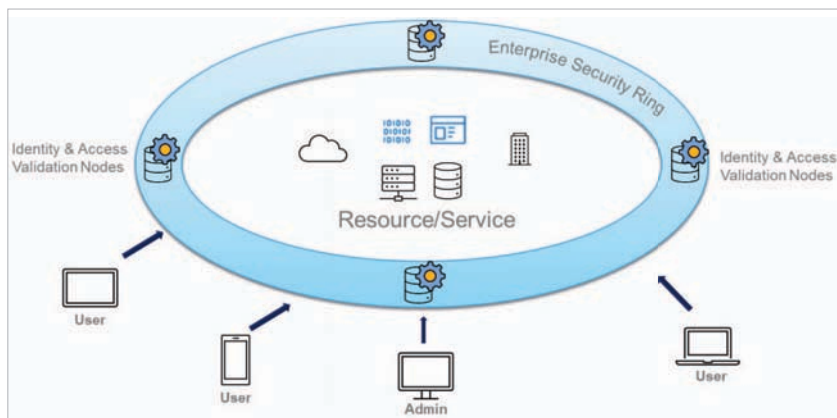


Figure 2: Blockchain defined enterprise security perimeter